Lab 3 — Finding a House with GIS

Final QGIS lab · Vector analysis · QGIS 3.44 LTR

Based on the QGIS Training Manual 3.44 (Vector Analysis lessons). Data: OpenStreetMap, from the Training Manual exercise data.

Goals of this lab

- Use the attribute table to choose features (filter).
- Understand why distances need a projected CRS, and save layers in one.
- Answer a real question with buffers, intersect and extract by location.

The data

- Unzip `Lab3_Data.zip` (or download the full dataset from `github.com/qgis/QGIS-Training-Data/archive/release_3.44.zip`).
- Put the folder `exercise_data` in a path with English letters only and no spaces, e.g. `C:\GIS\exercise_data`.
- Screenshots are from QGIS with this data. QGIS picks random colours, so your colours may differ.

Part 1 · The Problem and the Data

1. The Problem

You are an estate agent. Your clients want a house in **Swellendam** (a town in the Western Cape, South Africa) that is:

#	Criterion
1	in Swellendam
2	within **1 km** of a school
3	larger than **100 m²**
4	within **50 m** of a main road
5	within **500 m** of a restaurant

The GIS process

#	Step	In this lab
1	State the problem	Part 1 — the five criteria
2	Get the data	Part 1 — load, read and filter the layers
3	Analyze the problem	Parts 2–3 — CRS, buffers, intersect, extract, building size
4	Present the results	End of Part 3 — the solution layer and the results table

Term	Meaning
Feature	One object in a layer (one road, one building) = one **row** in the attribute table.
Field	One piece of information about every feature = one **column**, e.g. `highway`, `name`.
GeoPackage (.gpkg)	One file that holds several layers.

Term	Meaning
Filter	A condition on a field. Features that fail it are hidden from the layer and from every tool.

2. Get the Data

1. **Project** → **New** (or **New Empty Project** on the start page).
2. In the **Browser** panel, find `exercise_data`, right-click it → **Add as a Favorite** (Figure 1). It now appears under **Favorites**.

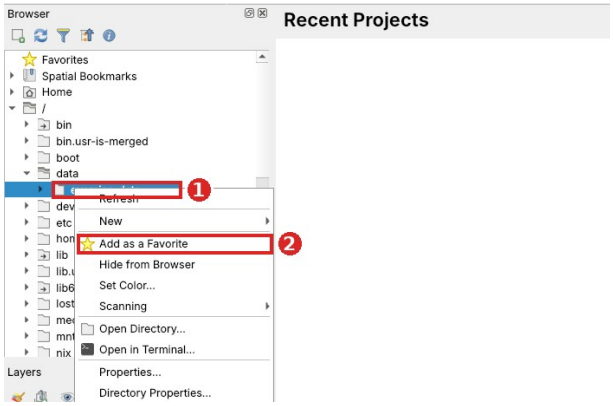


Figure 1 — Right-click the folder → Add as a Favorite

3. Expand `training_data.gpkg` and drag `buildings`, `roads`, `restaurants` and `schools` onto the map. Expand `landuse.sqlite` and drag `landuse` too.
4. Order the layers: `schools`, `restaurants`, `roads`, `buildings`, `landuse` — polygons at the bottom so they do not hide the rest.
5. Right-click each layer → **Show Feature Count** (Figure 2, number 1).
6. Double-click `schools` → **Symbology** → choose **red**; do the same for `restaurants` with **orange**, so they are easy to see.
7. Zoom in on the town and **Project** → **Save As…** → `lab3.qgz`.

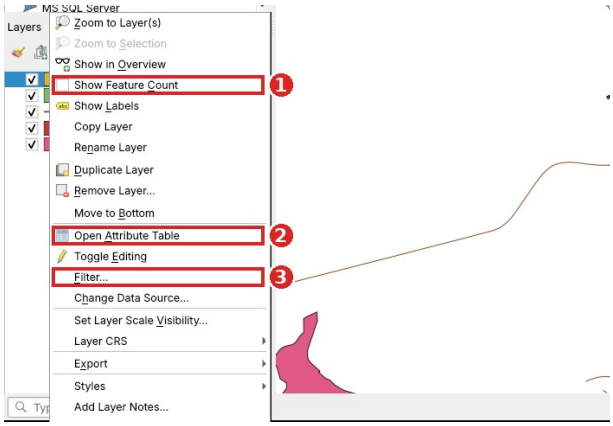


Figure 2 — The layer menu: (1) Show Feature Count, (2) Open Attribute Table, (3) Filter…

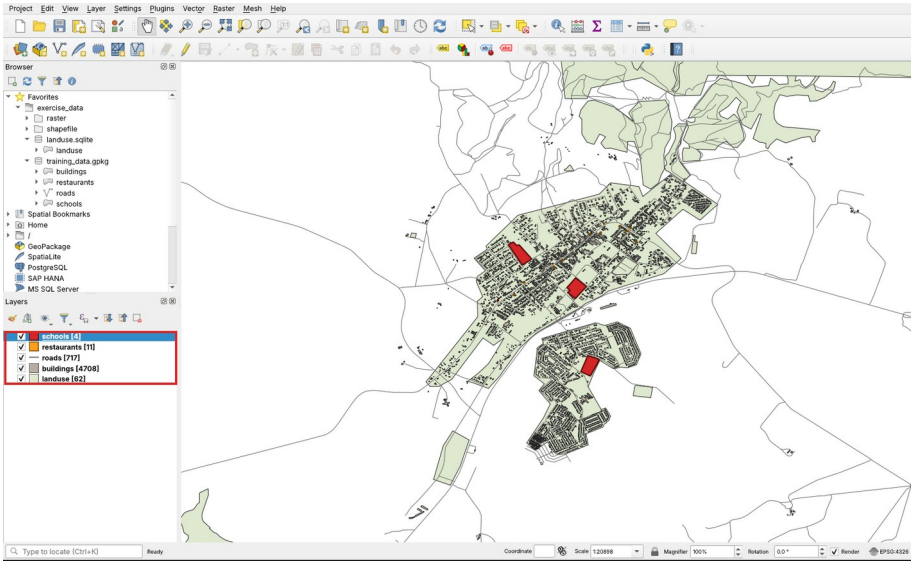
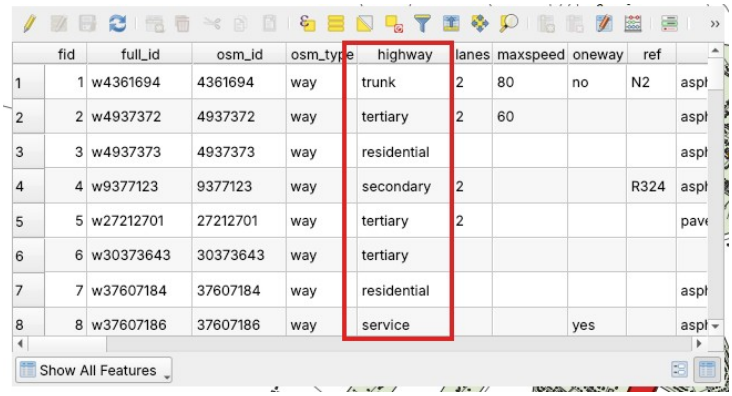


Figure 3 — The five layers with their feature counts

> ✓ **Check** schools [4], restaurants [11], roads [717], buildings [4708], landuse [62].

3. Read the Data — the Attribute Table

Right-click `roads` → **Open Attribute Table**. There are **717** rows, one per road. The field **highway** is the road type:



	fid	full_id	osm_id	osm_type	highway	lanes	maxspeed	oneway	ref	
1	1	w4361694	4361694	way	trunk	2	80	no	N2	aspl
2	2	w4937372	4937372	way	tertiary	2	60			aspl
3	3	w4937373	4937373	way	residential					aspl
4	4	w9377123	9377123	way	secondary	2			R324	aspl
5	5	w27212701	27212701	way	tertiary	2				pave
6	6	w30373643	30373643	way	tertiary					
7	7	w37607184	37607184	way	residential					aspl
8	8	w37607186	37607186	way	service			yes		aspl

Show All Features

Figure 4 — The roads table; highway is the road type

highway	Roads	highway	Roads
residential	268	path	32
track	133	primary	12
service	87	footway	10
unclassified	76	trunk_link	5
trunk	57	secondary	2
tertiary	34	road	1

Open the table of `schools` and scroll to `name`: the four schools that criterion 2 is measured from.

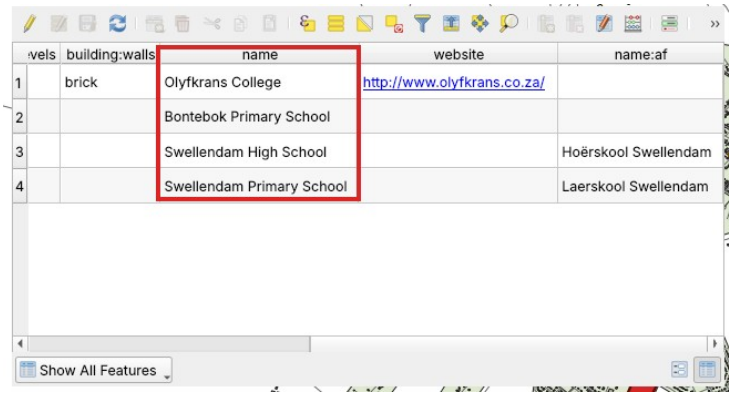


Figure 5 — The four schools

4. Keep Only the Main Roads — a Filter

Footways, paths, unclassified roads and tracks are not main roads, and some roads have no type at all (**NULL**). We hide them with a filter on `highway`.

1. Right-click `roads` → **Filter…** (Figure 2, number 3).
2. Double-click `highway` in **Fields** to start the expression, then complete it (Figure 6, number 2):
   `"highway" NOT IN ('footway', 'path', 'unclassified', 'track') AND "highway" IS NOT NULL`
3. Click **OK** (number 3).

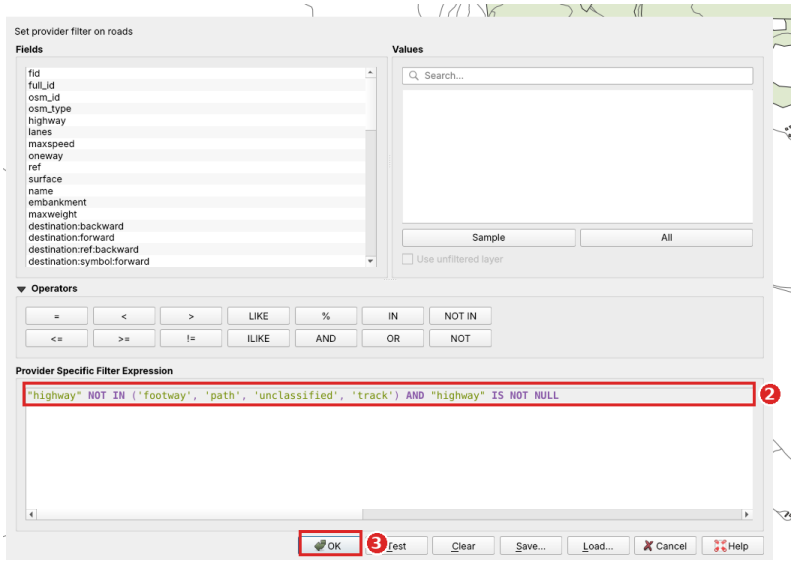


Figure 6 — The filter on roads

Part	Meaning
`"highway"`	The field — field names in double quotes.
`NOT IN ('footway', 'path', 'unclassified', 'track')`	Excludes these road types — text values in single quotes.
`AND "highway" IS NOT NULL`	…and the road must have a type.

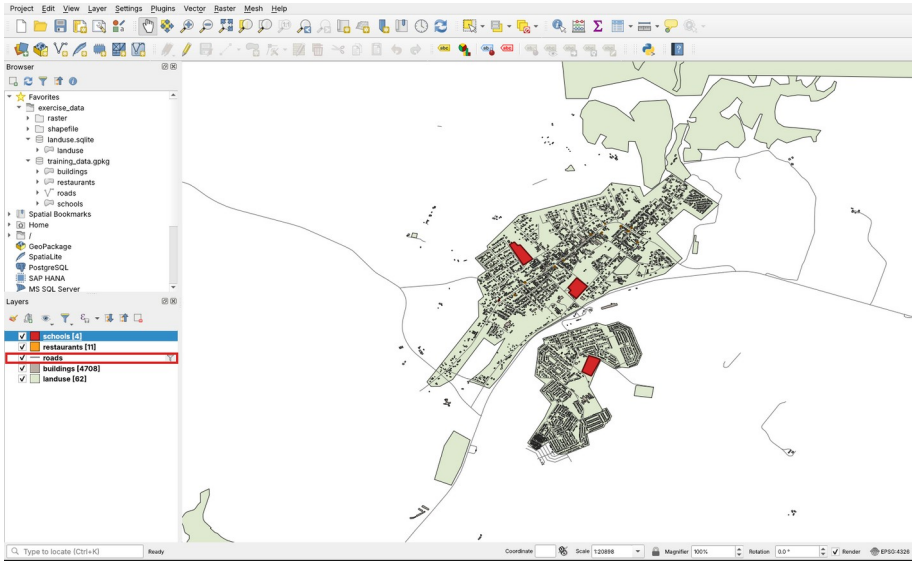


Figure 7 — A filter icon appears next to roads; tracks and paths are gone

- **466** of the 717 roads remain. Every tool from now on works only on these roads.
- To remove a filter: right-click → **Filter…** → **Clear** → **OK**.

Part 2 · The Coordinate Reference System

5. Why We Must Change the CRS

The data is in **WGS 84 (EPSG:4326)**, a geographic CRS: positions are in degrees of latitude and longitude, measured on a round Earth but drawn on a flat screen. Because of that, the scale is not the same everywhere on the map, and a degree is not a fixed distance.



Figure 8 — EPSG:4326: coordinates in degrees (latitude, longitude)

A **projected CRS** flattens the Earth in a controlled way and uses **metres**. Our criteria are distances (50 m, 1 km, 500 m) and an area (100 m²), so we need one. For Swellendam we use **WGS 84 / UTM zone 34S (EPSG:32734)**; for another region, choose the UTM zone of that region (Cairo, for example, is in zone 36N).

6. Change the Project CRS

QGIS reprojects “on the fly”: it can draw every layer in the CRS of the project, whatever CRS the layer is stored in.

1. Click the **EPSG:4326** button in the bottom-right corner. The CRS page opens (Figure 9, number 1).
2. Type `34S` in **Filter** (number 2).
3. Select **WGS 84 / UTM zone 34S — EPSG:32734** (number 3) and click **OK**.

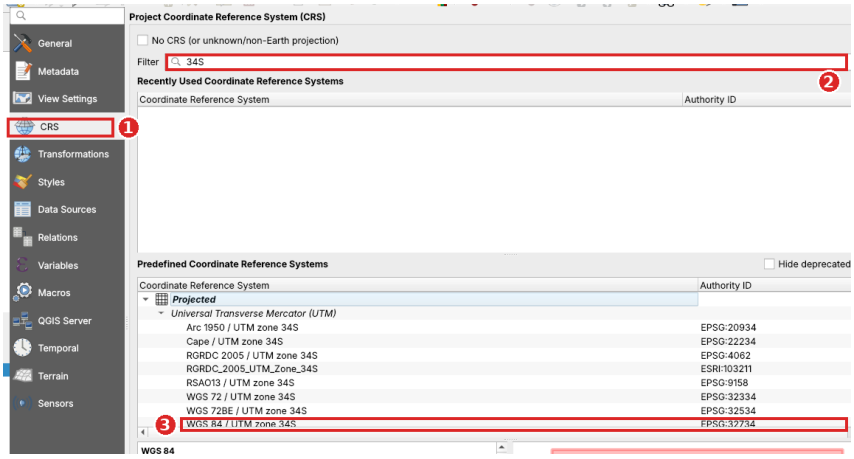


Figure 9 — Project CRS: filter “34S”, select WGS 84 / UTM zone 34S



Figure 10 — EPSG:32734: coordinates in metres

> **Project CRS and layer CRS are different things**
>
> - On-the-fly reprojection belongs to the **project**. A layer can be stored in another CRS and still appear in the right place.
> - Hover over a layer in the Layers panel: it is still in EPSG:4326. The analysis tools use the **layer** CRS, so we must save the layers in EPSG:32734.

7. Save Each Layer in EPSG:32734

1. Right-click `roads` → **Export** → **Save Features As…** (Figure 11).



Figure 11 — Export → Save Features As…

2. **Format:** GeoPackage (Figure 12, number 1).
3. **File name:** click **…** and name the new GeoPackage `vector_analysis` (number 2).
4. **Layer name:** `roads_34S` (number 3).
5. **CRS:** WGS 84 / UTM zone 34S — it is in the list as **Project CRS: EPSG:32734** (number 4). Click **OK**.

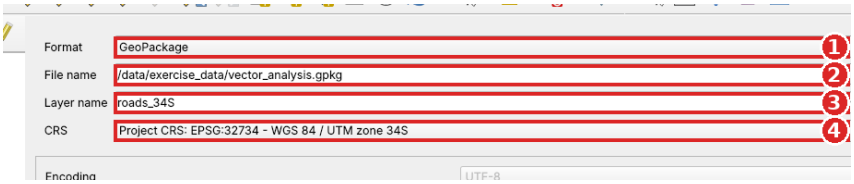


Figure 12 — Save Vector Layer As

6. Repeat for `buildings`, `restaurants`, `schools` and `landuse`, into the same `vector_analysis.gpkg`, adding `_34S` to each name. Each new layer is added next to the others in the GeoPackage.
7. Remove the old layers, then right-click any layer → **Zoom to Layer(s)**.

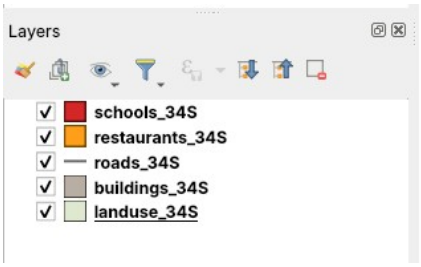


Figure 13 — The new layers, all in EPSG:32734

> ✓ **Check** Five layers ending in _34S. roads_34S has 466 features — the filter was kept, so only main roads were saved.

Part 3 · Analyze the Problem

8. The Processing Toolbox and the Buffer

1. Show only `roads_34S` and `buildings_34S`.
2. **Processing → Toolbox.** Almost every analysis tool of QGIS is here.
3. Type `buffer` in the search box (Figure 14, number 1) and double-click **Buffer** (number 2).

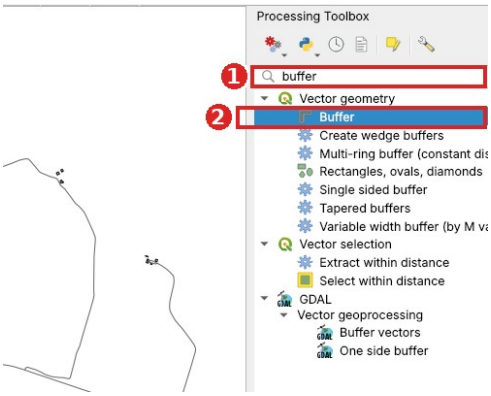


Figure 14 — Processing Toolbox: Buffer

What is a buffer?

- The area within a given distance of a feature. A 50 m buffer around a road is every place less than 50 m from it.

9. Within 50 m of a Main Road

1. **Input layer:** `roads_34S` (Figure 15, number 1).
2. **Distance:** `50` (number 2). The unit is metres because the layer is in a projected CRS. On a layer in degrees, QGIS warns you and suggests reprojecting it first.
3. Output: **…** → **Save to GeoPackage…** → `vector_analysis.gpkg`, layer name `roads_buffer_50m` (number 3). Click **Run** (number 4).

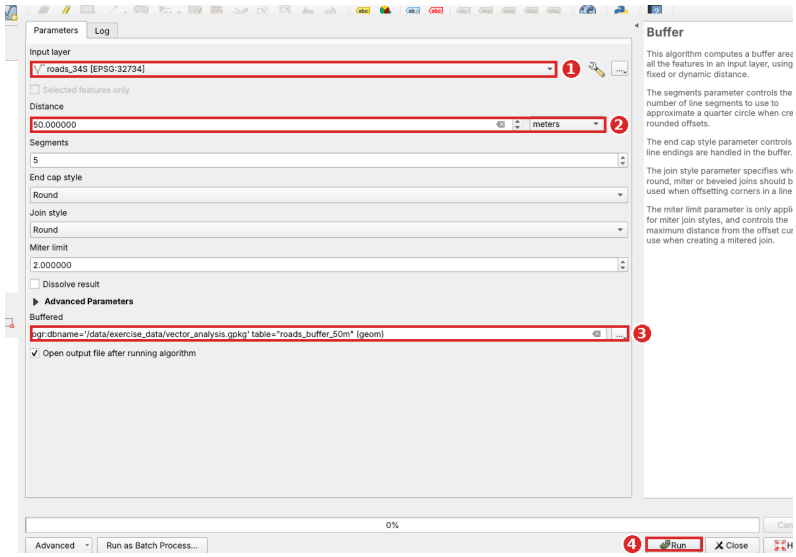


Figure 15 — Buffer: roads_34S, 50 m



Figure 16 — One separate buffer for each road (466 features)

Each road has its own buffer. Run **Buffer** again with **Dissolve result** checked and save it as `roads_buffer_50m_dissolved`: the result is one single area.

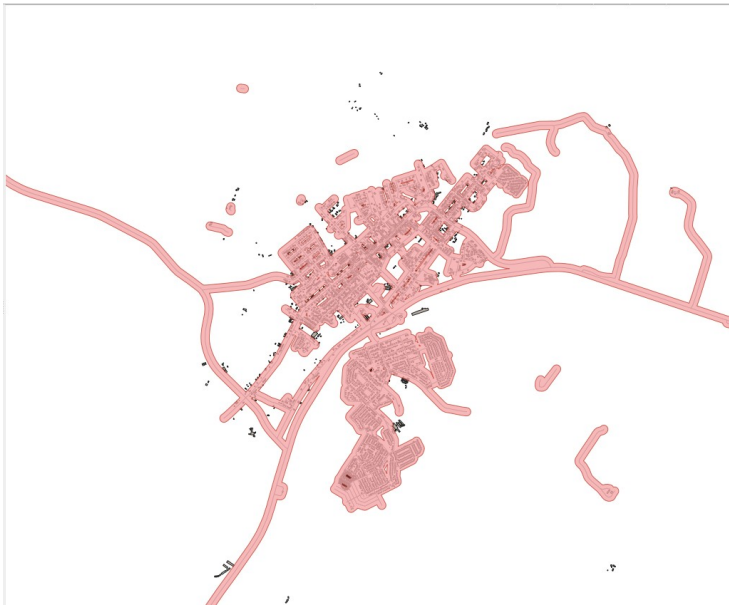


Figure 17 — The dissolved buffer: one area

10. Within 1 km of a School

1. **Buffer** with **Input layer** `schools_34S` (Figure 18, number 1).
2. **Distance:** 1 km — `1000` metres or `1` kilometre (number 2).
3. **Segments:** `20` (number 3) — smoother circles.
4. **Dissolve result** checked (number 4). Save as `schools_buffer_1km_dissolved` (number 5) and **Run**.

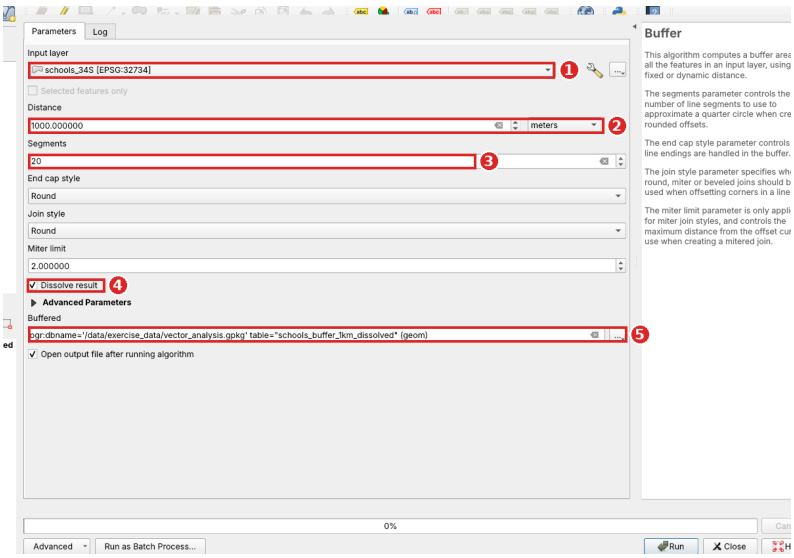


Figure 18 — Buffer: schools_34S, 1 km

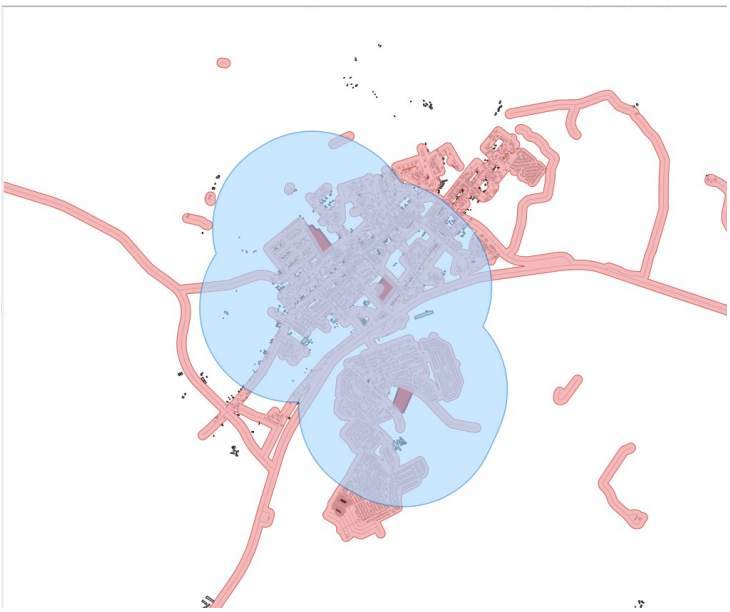


Figure 19 — 1 km around the schools (blue) and 50 m around the main roads (pink)

Straight line, not by road

- The 1 km is measured in a straight line from the school, not along the roads.

11. Both Criteria — Intersect

We want only the places that are inside **both** buffers. **Intersect** (group **Vector overlay**) keeps the overlapping part.

1. **Input layer:** `roads_buffer_50m_dissolved`; **Overlay layer:** `schools_buffer_1km_dissolved` (Figure 20).
2. Save as `road_school_buffers_intersect` and **Run**.

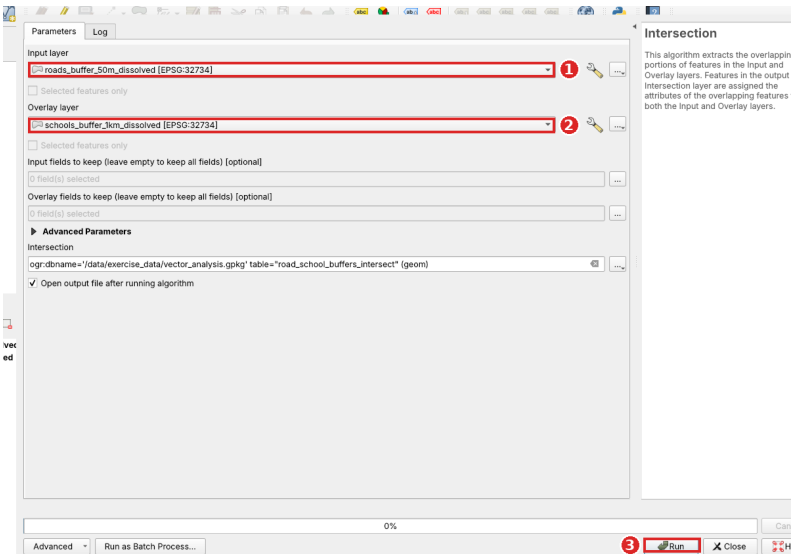


Figure 20 — Intersect

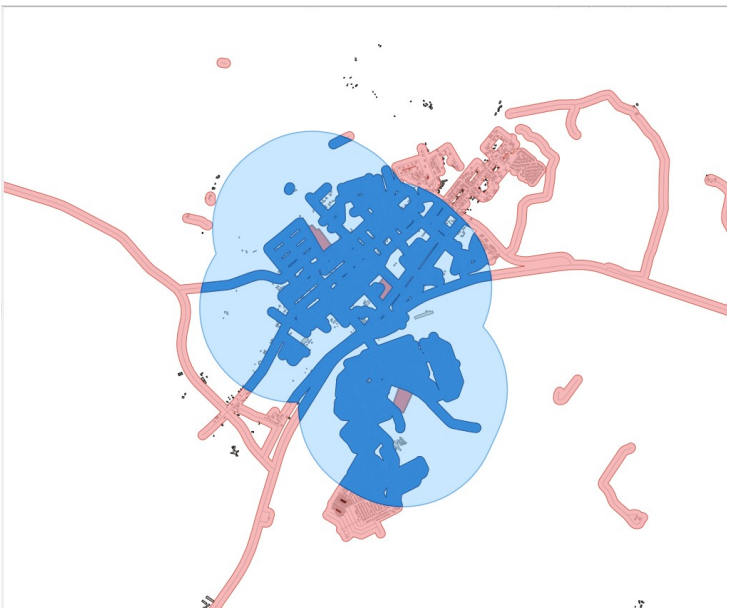


Figure 21 — Dark blue: within 1 km of a school and 50 m of a main road

You can now remove the two buffer layers and keep only the intersection.

12. Extract the Buildings

1. **Vector selection → Extract by location.**
2. **Extract features from:** `buildings_34S` (Figure 22, number 1). **Geometric predicate: intersect** (number 2).
3. **By comparing to the features from:** `road_school_buffers_intersect` (number 3).
4. Save as `well_located_houses` (number 4) and **Run**. Move it to the top of the Layers panel.

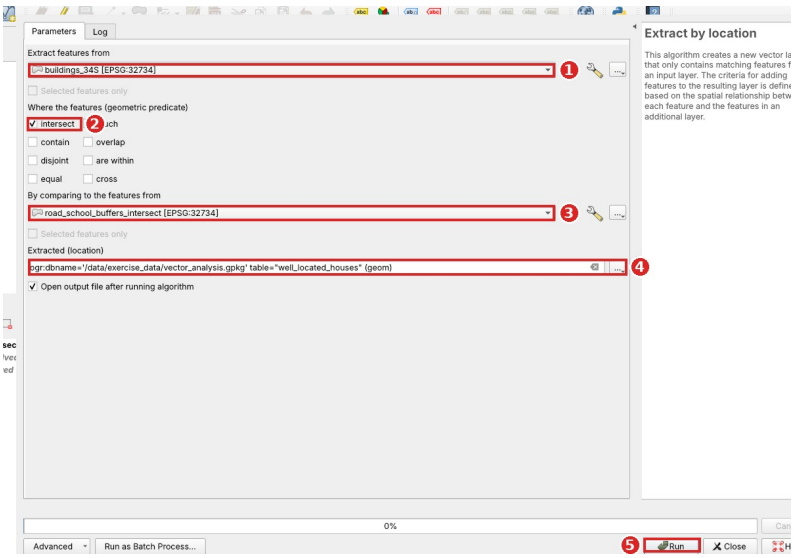


Figure 22 — Extract by location

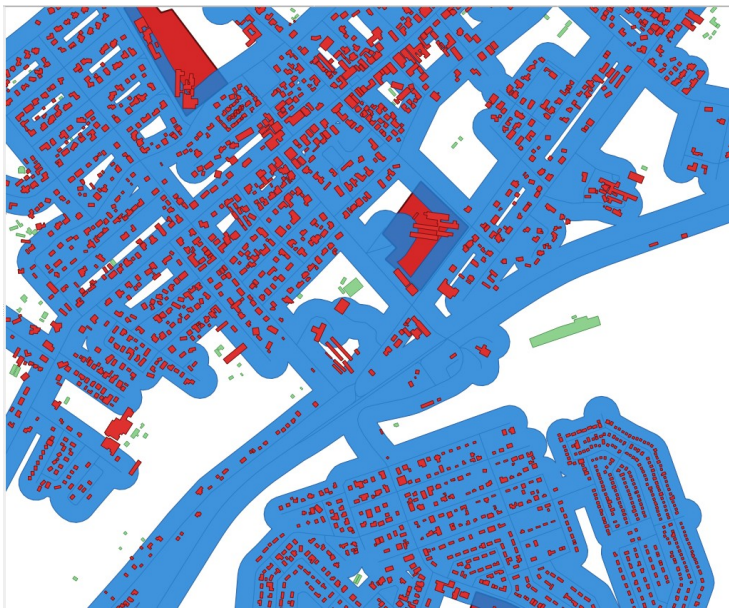


Figure 23 — Red: buildings that match; green: buildings that do not

3296 of the 4708 buildings are within 1 km of a school and 50 m of a main road.

13. Within 500 m of a Restaurant

1. **Buffer** `restaurants_34S` (Figure 24, number 1): **Distance** `500` (number 2), **Dissolve result** checked (number 3). Keep the default **Segments**. Save as `restaurants_buffer_500m_dissolved` (number 4) and **Run**.

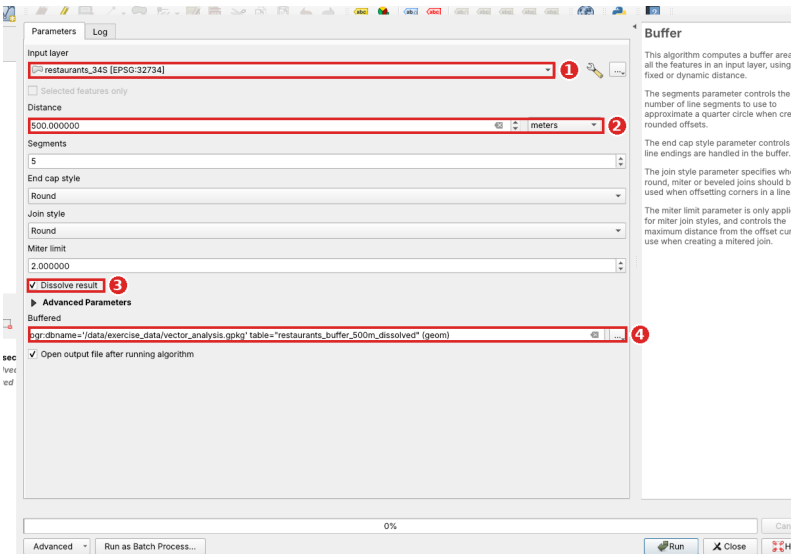


Figure 24 — Buffer: restaurants_34S, 500 m, dissolved



Figure 25 — 500 m around the restaurants (orange) over the well-located houses

2. **Extract by location:** extract from `well_located_houses` (Figure 26, number 1) the features that **intersect** (number 2) `restaurants_buffer_500m_dissolved` (number 3). Save as `houses_restaurants_500m` (number 4) and **Run**.

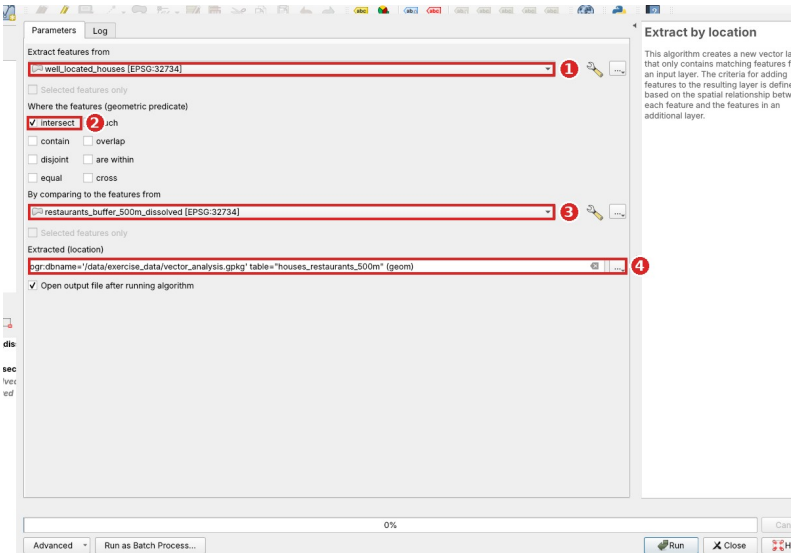


Figure 26 — Extract by location from well_located_houses



Figure 27 — Purple: houses that also meet the restaurant criterion

1247 houses meet criteria 2, 4 and 5.

14. Larger Than 100 m²

To keep the houses larger than 100 m² we first need the area of each building.

14a. Calculate the area of each building

1. In the **Layers** panel, click `houses_restaurants_500m` once to select it.
2. Click **Open Field Calculator** in the toolbar (the abacus icon). You can also open it from the toolbar of the attribute table.
3. Check **Create a new field** (Figure 28, number 1).
4. **Output field name:** `AREA` (number 2). **Output field type: Decimal number (real)** (number 3).
5. In the **Expression** box, type `$area` (number 4). `$area` is a ready-made QGIS function that gives the area of each feature in square metres; you can also double-click it in the **Geometry** group (number 5).
6. Click **OK**. The layer is now in editing mode (a pencil icon appears next to it).
7. Click **Toggle Editing** (the yellow pencil in the toolbar) and choose **Save**. The pencil next to the layer disappears.

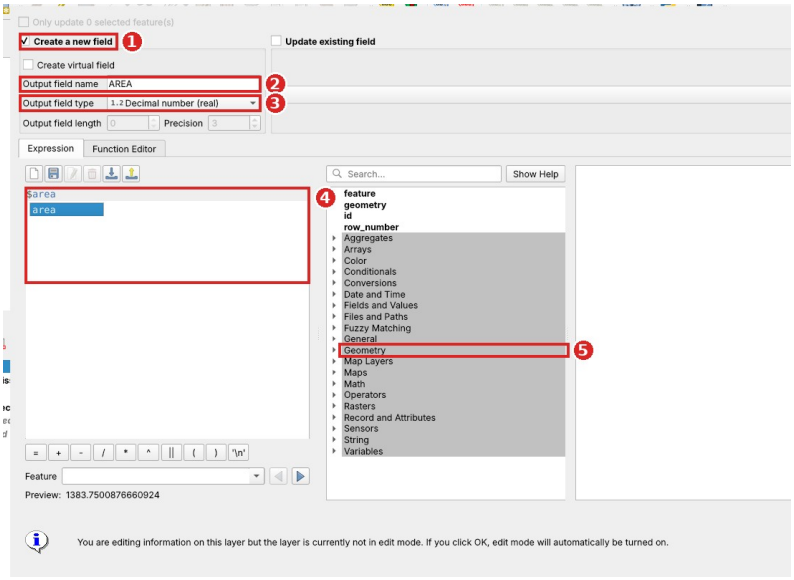


Figure 28 — Field Calculator: AREA = $area

The new field `AREA` is at the end of the attribute table:

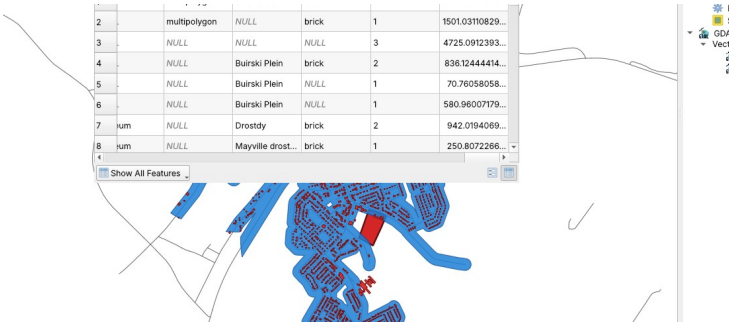


Figure 29 — The AREA field at the end of the table

14b. Keep only the buildings larger than 100 m²

Stop editing first (step 7): while the layer is being edited, the filter cannot be set.

1. Double-click `houses_restaurants_500m` → **Source** (Figure 30, number 1).
2. Under **Provider Feature Filter**, click **Query Builder** (bottom right). The box above it only shows the filter; it cannot be typed in.
3. In the Query Builder, type `"AREA" >= 100` and click **OK**.
4. Back in **Source**, the filter appears in the box (number 2). Click **OK**.

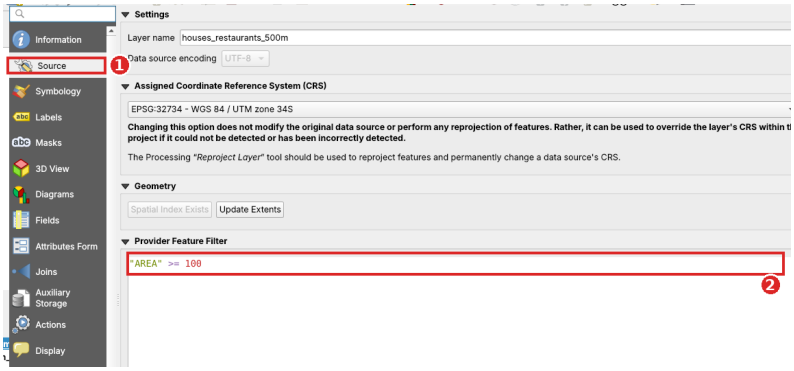


Figure 30 — Source → Provider Feature Filter



Figure 31 — The houses that meet all five criteria

996 houses meet all five criteria. Save them as a new layer: right-click → **Export** → **Save Features As…** → `vector_analysis.gpkg`, layer name `solution`.

Step	Layer	Features
All buildings	buildings_34S	4708
1 km from a school and 50 m from a main road	well_located_houses	3296
… and 500 m from a restaurant	houses_restaurants_500m	1247
… and larger than 100 m²	solution	**996**

References

- QGIS Training Manual — Vector Analysis — https://docs.qgis.org/3.44/en/docs/training_manual/vector_analysis/basic_analysis.html
- QGIS Training Manual — Reprojecting and Transforming Data — https://docs.qgis.org/3.44/en/docs/training_manual/vector_analysis/reproject_transform.html
- Exercise data — https://github.com/qgis/QGIS-Training-Data